

Abhishek Kumar Singh

Senior Site Reliability/DevOps Engineer

abhishek.techie21@gmail.com | +91-7411300272 | Bengaluru, India | linkedin.com/in/abhishek-techie21

PROFILE

Senior SRE / DevOps Engineer with 7+ years leading production reliability and cloud platforms across AWS, Azure, and GCP. Expert in Kubernetes, Terraform, Ansible, CI/CD, and large-scale observability systems. Proven record reducing MTTR, rollout risk, and cloud spend through automation, progressive delivery, and AIOps. Driving platform maturity through SLOS, release governance, and predictive capacity planning.

WORK EXPERIENCE

SLK Software, Team Leader

01/2025 – Present | Bengaluru

Client: Emerson, Fifth Third Bank, HNB bank | Emerson

- Built end-to-end observability for Azure workloads using Managed Prometheus, Grafana, and Azure Monitor, reducing MTTR by 25% and false alerts by 30%.
- Designed monitoring for Azure Kubernetes Service (AKS), PostgreSQL, and HTTP endpoints with Service Level Objective (SLO) and Service Level Agreement (SLA) dashboards and actionable alerting.
- Delivered infrastructure changes through Infrastructure as Code (IaC) using Terraform and Ansible with standardized modules, approval workflows, and policy guardrails to improve release safety and compliance.
- Partnered with product owners and platform engineering teams to define golden signals and production-readiness checklists for new services.

Fifth Third Bank

- Automated operating system patching and application deployments using Ansible and Jenkins, reducing manual effort by 70%.
- Enabled event correlation and auto-baselining in Dynatrace, reducing P1/P2 incidents by ~20%.
- Introduced runbooks and incident-response playbooks integrated with PagerDuty and Opsgenie, improving on-call consistency and time to mitigation.
- Acted as a Build-and-Release-focused Site Reliability Engineer, owning CI/CD pipelines and production release workflows across Kubernetes platforms.
- Designed and enforced release Service Level Objectives (SLOs) and post-deployment health checks using Prometheus, Grafana, Dynatrace, and Splunk, reducing rollback frequency and improving MTTR by 25-40%.
- Implemented blue-green and canary deployments on auto-scaling infrastructure integrated with Jenkins.

HNB Bank

- Orchestrated event-driven remediation using PowerShell and Ansible triggered by anomaly signals, reducing manual toil by 30%.
- Applied operating system hardening and patch compliance for cloud and on-premises nodes aligned to CIS benchmarks, improving audit scores.
- Expanded Dynatrace and Splunk dashboards with threshold tuning and business key performance indicators (KPIs), improving the signal-to-noise ratio for alerting.
- Designed Google Cloud Platform (GCP) environments aligned with Google Cloud Architecture Framework, leveraging Compute Engine, Google Kubernetes Engine (GKE), Cloud Load Balancing (HTTP/S & Internal), Cloud Armor (WAF), Cloud CDN, IAM, Service Accounts, Workload Identity, Cloud KMS, and Secret Manager and Cloud Storage (GCS), using Infrastructure as Code with Terraform.
- Designed secure GCP networking architectures using VPC, private subnets, Shared VPC, VPC Peering, Cloud NAT, and Firewall Rules, enabling private service communication and controlled egress for Tier-1 workloads.
- Managed Google Kubernetes Engine (GKE) deployments with Horizontal Pod Autoscaler (HPA) and Vertical Pod Autoscaler (VPA), resource governance, and secure ingress and egress, improving platform stability and cost efficiency.

Impetus Technologies, SRE/Devops Engineer

06/2021 – 12/2024 | Noida

Client: Transunion |

- Migrated legacy monoliths to Kubernetes microservices using Docker, Google Kubernetes Engine (GKE), and Amazon Elastic Kubernetes Service (EKS), improving scalability, resilience, and release velocity.
- Designed Amazon Web Services (AWS) environments from the ground up aligned with the AWS Well-Architected Framework, leveraging EC2, Auto Scaling Groups (ASG), Application Load Balancers (ALB), Web Application Firewall (WAF), CloudFront, S3, and RDS, with Infrastructure as Code using Terraform and CloudFormation.
- Designed cloud networking and security architectures across AWS and GCP, including Virtual Private Cloud (VPC) and Virtual Network (VNet) with private subnets, secure ingress via ALB and Application Gateway, and controlled egress using NAT Gateways and firewalls, eliminating public exposure for Tier-1 services.
- Implemented Identity and Access Management (IAM), Role-Based Access Control (RBAC), and secrets management using AWS Key Management Service (KMS) and Google Secret Manager, integrated with Continuous Integration / Continuous Deployment (CI/CD) pipelines to enforce least privilege and automated credential rotation.
- Implemented production-grade CI/CD pipelines using Jenkins, incorporating blue-green and canary deployments, reducing rollback frequency by 30%.
- Developed Ansible playbooks to automate software provisioning, deployment, configuration updates, and system maintenance, reducing manual intervention and operational toil.
- Standardized Terraform modules for network, compute, and Kubernetes add-ons, cutting provisioning time by 80% and improving reproducibility across environments.
- Leveraged Splunk and the ELK Stack for log analytics with custom dashboards, accelerating troubleshooting and reducing mean time to detect (MTTD).
- Conducted vulnerability assessments and patching programs to mitigate security risks and maintain regulatory and audit compliance.
- Utilized Prometheus, Grafana, and Dynatrace for real-time monitoring and alerting, enabling proactive system management and performance optimization.
- Configured and deployed secure VPN solutions using Libreswan and StrongSwan for enterprise customers.

Cavisson Systems, Software Engineer

10/2018 – 05/2021 | Noida

Client: Kohl's |

- Implemented robust monitoring, alerting, and observability solutions using APM, Prometheus, Grafana, and logs to reduce downtime for critical applications.
- Conducted performance testing, diagnostics, root cause analysis (RCA), and optimization to improve application reliability, throughput, and system efficiency.

- Designed and maintained Jenkins CI/CD pipelines for automated build, test, and deployment; integrated with Git, Maven, and Nexus to improve release cadence and reduce errors.
- Provisioned and managed cloud infrastructure (AWS and GCP using Terraform and Ansible, established logging, monitoring baselines, and monthly patch workflows.
- Containerized workloads using Docker, Docker Compose, and Kubernetes, streamlining deployment, scaling, and environment consistency.
- Collaborated with development and QA teams to design scalable, high-performance, and resilient systems that met client functionality, reliability, and SLA/SLO requirements.
- Implemented automation and scripting (Python, Bash) for deployment, monitoring, and incident remediation, reducing manual intervention and MTTR.
- Applied infrastructure-as-code (IaC) best practices and GitOps principles to ensure reproducible, version-controlled, and auditable deployments.

TOOLS

AIOps & Observability AI: Dynatrace Davis AI (Anomaly Detection, Root Cause Analysis) | Predictive capacity analytics (CPU, memory, pod saturation forecasting)

AI / LLM Skills: Prompt Engineering, Llama /open-source LLMS, Agentic AI workflows, Fast API, LlamaIndex

Operating Systems: Linux, Windows

Servers & Web: NGINX, Apache HTTP, Apache Tomcat, JBoss

Cloud Services: AWS, GCP, Azure

Configuration & Infrastructure Management: Ansible, Ansible Tower, Terraform

CI/CD Tools: Jenkins, Harness CI/CD, Argo CD

Container Technology: Docker, Docker-Compose, Kubernetes, Helm

Database: PostgreSQL, MySQL, RDS

Monitoring & Logging Tools: Grafana, Prometheus, Cortex, Fluentd, File beat, Net diagnostic, Splunk, ELK Stack, Dynatrace, Google Cloud Monitoring, Amazon CloudWatch, Thousand Eyes, AI observability

Security & Vulnerability: SonarQube, Rapid7 Nexpose, CrowdStrike Falcon

Ticketing Tools: JSM, ServiceNow

Version Control: GitHub, GitLab

Incident Management & Automation: Pager Duty, Opsgenie, StackStorm, Shell scripting, AWX

Artifact & Image Management: Docker Trusted Registry (DTR), Harbor, Nexus

Messaging & Streaming: RabbitMQ, ActiveMQ, Kafka

Documentation & Collaboration: Confluence, SharePoint, Google Workspace, Microsoft Teams + OneNote

VPN Solutions: Libreswan, StrongSwan

SPECIALIZED SKILLS

- Acted as Lead SRE for public-facing production systems, managing incident response, client communications, and promoting DevOps best practices.
- Diagnosed Java performance issues including memory leaks, GC pauses, and slow endpoints; monitored JVM health metrics such as heap usage, GC, thread count, and response times.
- Executed performance tuning and optimization across applications and infrastructure to ensure SLA compliance.
- Designed and tested disaster recovery (DR) and high availability (HA) strategies for enterprise-grade applications.
- Managed configuration, deployment, and automation using Ansible and Infrastructure-as-Code principles.
- Implemented monitoring, logging, and observability frameworks across applications and cloud environments using Prometheus, Grafana, and ELK Stack.
- Built and maintained CI/CD pipelines to automate build, test, and deployment; integrated with Git, Jenkins, and container registries.
- Led migration of monolithic applications to containerized and orchestrated environments using Docker and Kubernetes.
- Ensured security and vulnerability management, performing audits, patching, and compliance checks.
- Coordinated ticketing, documentation, and collaboration using Jira, Confluence, and Slack for cross-team visibility.
- Provided on-call production support, troubleshooting complex incidents and maintaining IT operations reliability.

ENGINEERING INITIATIVES

- Designed and built a web based Terraform dashboard integrated with an AI chatbot assistant, streamlining infrastructure provisioning and improving user experience.
- Leveraged open-source LLMS (Llama) for internal SRE knowledge management and root cause analysis (RCA) exploration.
- Led POC for SLK Direct, aggregating data from Dynatrace and Service Now to create unified operational views and workflows, accelerating SRE adoption.
- Integrated AWX for event automation and self-service operations, reducing manual interventions across incident management and change processes.
- Built POCs with Harness (CI/CD) demonstrating automated build, deploy, progressive delivery, and policy enforcement workflows.
- Evaluated Nagios, Thousand Eyes, and New Relic for infrastructure, network, and application performance monitoring; produced comparative analyses and adoption recommendations.

EDUCATION

Bachelor of Technology (B.Tech) in Electronics, *Ideal Institute of Technology*
Percentage: 73%

08/2014 – 07/2018 | Ghaziabad